81. The 'Common Carbon Metric', supported by UNEP, has been developed for?

(a) assessing the carbon footprint of building operations around the world
(b) enabling commercial farming entities around the world to enter carbon emission trading
(c) enabling governments to assess the overall carbon footprint caused by their countries
(d) assessing the overall carbon footprint caused by the use of fossil fuels by the world in a unit time

Ans. (a): The 'common carbon Metric', supported by UNEP has been developed for assessing the carbon foot print of building operations around the world.
Hence, option (a) is the correct answer.

82. Which of the following have species that can establish symbiotic relationship with other organisms?

1. Cnidarians
2. Fungi
3. Protozoas

Select the correct answer using the code given below.

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d): Symbiosis is a close ecological relationship between the individuals of two (or more) different species. The relationship between cnidarians and dinofl agellate algae is termed as "symbiotic". Two common mutualistic relationships involving fungi are mycorrhiza and lichen. Symbiosis in protozoa mostly represents a close mutualistic association between a protozoan and unicellular symbionts (bacteria, cyanobacteria or/and unicellular algae) or protozoans and a multicellular organism (ruminants, lower termites, wood-eating cockroaches, plants).

83. How is permaculture farming different from conventional chemical farming?

1. Permaculture farming discourages monocultural practices but in conventional chemical farming, monoculture practices are predominant.
2. Conventional chemical farming can cause increase in soil salinity but the occurrence of such phenomenon is not observed in permaculture farming.
3. Conventional chemical farming is easily possible in semi-arid regions but permaculture farming is not so easily possible in such rgions.
4. Practice of mulching is very important in permaculture farming but not necessarily so in conventional chemical farming.

Select the correct answer using the code given below.

(a) 1 and 3 (b) 1, 2 and 4
(c) 4 only (d) 2 and 3

Ans. (b) : Since the principles of permaculture discourage monoculture, it opens up the opportunity for growing a wide variety of grains, fruits and vegetables, and widens one's food basket. It can be possible in all kinds of climate including semi-arid regions. Hence, Statement-3 is not correct.
Permaculture cultivation discourages monoculture practices, improves soil fertility and encourages mulching. Hence, Statements 1, 2 and 4 are true.
Hence option (b) is the correct answer.

84. With reference to 'palm oil', consider the following statements:

1. The palm oil tree is native to Southeast Asia.
2. The palm oil is a raw material for some industries producing lipstick and perfumes.
3. The palm oil can be used to produce biodiesel.

Which of the statements given above are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (b): Palm oil is native to Africa and brought to southeast Asia during colonial times, palm oil plantations now cover huge areas of Malaysia and Indonesia.
Palm oil is used in lipstick as it holds color well, doesn't melt at high temperatures, and has a smooth application and virtually no taste.
Palm can be used for making biodiesel also.
Hence, Statement-1 is not correct while Statement-2 and 3 are correct.

85. Consider the following statements :

1. The Global Ocean Commission grants licenses for seabed exploration and mining in international waters.
2. India has received licenses for seabed mineral exploration in international waters.
3. 'Rare earth minerals' are present on seafloor in international waters.

Which of the statements given above is/are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (b) : Statement – 1 (Not – Correct) – The Global Ocean Commission was international initiative between 2013 and 2016 to raise awareness and promote action to address, the degradation of the ocean and help restore it to full health and productivity. Licence for Seabed exploration and mining in international waters are granted by International Seabed Authority.
Statement – 2 (Correct) – India has been exploring deep sea for minerals since 1981. In 2017, India's exclusive rights to explore polymetallic nodules from seabed in central Indian Ocean Basin have been extended by five years.
Statement – 3 (Correct) – The deep seabed contains two potential sources for rare earth elements, polymetallic nodules which typically contain, manganese, nickel, cobalt, copper and rare earth minerals.
Hence option (b) is the correct answer.

86. In the context of India's preparation for Climate-Smart Agriculture, consider the following statements :

1. The 'Climate-Smart Village' approach in India is a part of a project led by the Climate change, Agriculture and Food Security (CCAFS), an international research programme.

2. The project of CCAFS is carried out under Consultative Group on International Agriculatural Research (CGIAR) headquartered in France.
3. The International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) in India is one of the CGIAR's research centres.

Which of the statements given above are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d): Statement 1 is correct. The Climate-Smart Village project in India is a program of CGIAR Research Program on Climate Change, Agriculture and Food Security (CCAFS). The CCAFS started piloting the Climate-Smart Village in 2012 in Africa (Burkina Faso, Ghana, Mali, Niger, Senegal, Kenya, Ethiopia, Tanzania, and Uganda) and South Asia (Bangladesh, India, and Nepal).
Statement 2 is correct. Climate Change, Agriculture and Food Security (CCFAS) is carried out under CGIAR (formerly the Consultative Group for International Agricultural Research). Headquarter of CGIAR is in Montpellier, France. CGIAR is a global partnership that unites international organizations engaged in research about food security.
Statement 3 is correct. International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) is a CGIAR Research Center. ICRISAT is a non-profit, non-political public international research organization that conducts agricultural research for development in Asia and sub-Saharan Africa with a wide array of partners throughout the world.
Hence, option (d) is the correct answer.

87. The vegetation of savannah consists of grassland with scattered small trees, but extensive areas have no trees. The forest development in such areas is generally kept in check by one or more or a combination of some conditions. Which of the following are such conditions?

1. Burrowing animals and termites
2. Fire
3. Grazing herbivores
4. Seasonal rainfall
5. Soil properties

Select the correct answer using the code given below.

(a) 1 and 2 (b) 4 and 5
(c) 2, 3 and 4 (d) 1, 3 and 5

Ans. (c) : Savannas are defined based on vegetation structure, the central concept being a discontinuous tree cover in a continuous grass under storey. Fire, grazing herbivore and seasonal rainfall are behind the scarce forest development in the region.
Hence, option (c) is the correct answer.

88. Consider the following statements:

1. Moringa (drumstick tree) is a leguminous evergreen tree.
2. Tamarind tree is endemic to South Asia.
3. In India, most of the tamarind is collected as minor forest produce.
4. India exports tamarind and seeds of moringa.
5. Seeds of moringa and tamarind can be used in the production of biofuels.

Which of the statements given above are correct?

(a) 1, 2, 4 and 5 (b) 3, 4 and 5
(c) 1, 3 and 4 (d) 1, 2, 3 and 5

Ans. (b) : Statement 1 is incorrect: Moringa Oleifera is a medium-sized evergreen tree that is native to Africa and Asia. Also known as the Moringa Tree, the Drumstick Tree or the Miracle Tree. This is non-leguminous tree.
Statement 2 is incorrect: Tamarind, (Tamarindus indica), evergreen tree of the pea family (Fabaceae), native to tropical Africa.
Hence, option (b) is the correct answer.

CURRENTAFFAIRS

89. With reference to casual workers employed in India, consider the following statements :

1. All casual workers are entitled for Employees Provident Fund coverage.
2. All casual workers are entitled for regular working hours and overtime payment.
3. The government can by a notification specify that an establishment or industry shall pay wages only through its bank account.

Which of the above statements are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d) : Statement – 1 (Correct) – According to Employees Provident Funds and Miscellaneous Provisions Act, An employee cannot differentiate between contractual and permanent employees and all causal workers are also entitled to social security benefits under the EPFO Act.
Statement – 2 (correct) – The Employees Provident Funds Act, 1952, provide Social Security to All Casual Workers are entitled for regular working hours and over time payment.
Statement – 3 (Correct) – According to the Payment of Wages Act, 1936, Government to specify the industrial or other establishment, by notification in the official Gazette, which enable the employers to pay wages to their employees by crediting to their bank account.
Hence, option (d) is the correct answer.

90. 'R2 Code of Practices' constitutes a tool available for promoting the adoption of

(a) environmentally responsible practices in electronics recycling industry
(b) ecological management of 'Wetlands of International Importance' under the Ramsar Convention
(c) sustainable practices in the cultivation of agricultural crops in degraded lands
(d) 'Environmental Impact Assessment' in the exploitation of natural resources